The Trapped-Ion Ultrametric Testbed: A Falsifiability Register for Testing p-Adic Structure in Quantum Dynamics

Rowan Brad Quni-Gudzinas

2026-08-19

Abstract

Sixteen published records from a single research program, spanning December 2025 to August 2026, are here organized into one testable claim: trapped-ion quantum simulators are the first near-term platform on which ultrametric (p-adic) structure in quantum dynamics can be accepted or rejected by measurement, and passive, dissipation-structured hardware is the engineering embodiment of that structure. The synthesis contributes a falsifiability register: five observables, each with a pre-registered prediction, a kill-condition, and an apparatus already demonstrated in the cited records. It also records what the program has already ruled out, including the finding that generic Page-Wootters clock-rest coupling violates ultrametricity at a 29-35 percent rate while diagonal coupling yields exact ultrametric structure – the very asymmetry the trapped-ion protocol measures – and the economic finding that trapped ions rank last among seventeen platforms on a joules-per-solution basis, making the ion trap a physics testbed rather than a production economics play. The paper discloses its premise depth: the falsifiable content rests on two named imported inputs, and the claims end where those inputs begin. A practitioner section specifies what can be built today without any new theory.

1. Introduction

Between December 2025 and August 2026, a research program published sixteen records on trapped-ion quantum computation, p-adic geometry, and the epistemology of physics. Read individually, they look like three different projects: an experimental physics effort, a mathematical physics effort, and a philosophy-of-science effort. This paper argues they are one project, and that reading them as one project changes what a practitioner should do next.

The unifying claim is structural, not metaphysical. A tree-shaped hierarchy – the Bruhat-Tits tree of p-adic geometry, the hierarchy of conditional quantum states under a global constraint, the hierarchy of stabilizer and dissipative error correction, and the hierarchy of claims ranked by evidential weight – appears in all three movements (QNFO Research and Quni-Gudzinas 2026a; Quni-Gudzinas 2026k, 2025, 2026l). The claim made here is that this appearance is testable on hardware that already exists, and that a set of falsifiable observables with kill-conditions is the correct way to carry the claim forward. This paper deliberately does NOT claim the tree structure has been confirmed. It claims only that the structure is now testable, and it supplies the register by which it will be tested.

Why a reader should care: near-term quantum hardware needs discriminating experiments, not only gate-count benchmarks. A measurement that could rule out an entire class of ultrametric interpretations of quantum dynamics – or fail to rule it out – is worth more than another demonstration that a simulator can simulate. This paper hands an experimental group a ready-to-run menu of such measurements, with predicted values, kill-conditions, and the apparatus requirements already worked out in the cited records.

The premise-depth disclosure is stated here and formalized in Section 10. The derived content of this paper is organizational: resource counts, predicted values, and the register itself. Two named imported inputs carry the physical content: the Ultrametric Bridge Theorem, which states that conditional quantum states under a global constraint necessarily form ultrametric hierarchies (Quni-Gudzinas 2026r), and the Silent Parameter Principle, which identifies the truncated SU(2) representation ring with the character ring of the 2-adic units (QNFO Research and Quni-Gudzinas 2026b). The premises END there: this paper neither re-derives nor independently verifies those inputs, and no claim in this paper is stronger than its inputs.

2. The Sixteen Records: Three Movements

The sixteen records sort into three movements.

The testability movement asks whether p-adic structure is measurable on trapped-ion hardware. A protocol for testing ultrametricity via a Page-Wootters clock experiment specifies a single trapped ion, laser-driven carrier and sideband transitions, conditional state tomography, and an eight-week timeline on existing apparatus (QNFO Research and Quni-Gudzinas 2026a). A companion analysis of 8,000 simulated Wheeler-DeWitt systems establishes the prediction the protocol measures: generic clock-rest coupling violates the Parisi ultrametricity condition at a 29-35 percent rate, while coupling diagonal in the clock eigenbasis yields exact ultrametric structure (Quni-Gudzinas 2026c). The Zitterbewegung is re-read as an adelic observable with a 2-adic frequency of $\sqrt{2} \cdot 2mc^2/\hbar$, a 41.4 percent deviation from the Archimedean value, and a single-ion timeline of 2028-2032 (QNFO Research and Quni-Gudzinas 2026b). Two feasibility studies examine whether a vortex-enhanced amplification mechanism can lift the sub-Compton-scale signal above the noise floor, and conclude the question is open pending signal-to-noise work (Quni-Gudzinas 2026u, 2026v). A due-diligence assessment of the Innsbruck qudit program verifies the measured light-shift gate fidelities (99.6 to 93.7 percent for dimensions 2 through 5) that such experiments would inherit (Quni-Gudzinas 2026t; Ringbauer et al. 2022; Hrmo et al. 2023; Meth et al. 2024). A Monte-Carlo study extracts effective transient dimensions of 6.2 and 7.9 for diffusion on p-regular trees with p equal to 2 and 3, against the integer-line baseline of 1.0 (Quni-Gudzinas 2026j). An audit of a tunable quantum neural network demonstrates the program's testing discipline: the strongest claim survives and the advantage claims do not (QNFO Research Collective and Quni-Gudzinas 2026).

The architectural movement asks what hardware built on the same structure looks like. A room-temperature trapped-ion architecture stabilizes Gottesman-Kitaev-Preskill states autonomously through engineered dissipation, converging to the GKP manifold when the cooling rate exceeds the heating rate by more than π (Quni-Gudzinas 2025). The Spin-Free Substrate protocol simulates Posner-molecule nuclear spin dynamics with six global Molmer-Sorensen gates per simulated second at 96.9 percent circuit fidelity, and names the design principle connecting Kane-type, Posner-type, and trapped-ion architectures (Quni-Gudzinas 2026p, 2026q). Quantum Architectonics argues the passive path – letting structure suppress error – against the active-correction path that fights the second law (Quni-Gudzinas 2026h). A p-adic metrology proposal claims sub-shot-noise sensing from hierarchical noise correlations on a room-temperature photonic platform with roughly ten entangled photons (Quni-Gudzinas 2026g). A dimensionless reformulation of fifty-three physics equations in Planck units supplies the place-democratic bookkeeping the movement uses (Quni-Gudzinas 2026n).

The methodological movement supplies the discipline. The Falsifiability Crisis record diagnoses five structural patterns by which contemporary fundamental physics has become effectively unfalsifiable, and proposes the Bayesian delta-log-odds gate as the remedy (Quni-Gudzinas 2026l). The Reification record names the failure mode the other two movements must avoid: mistaking a mathematical model for a mind-independent object (Quni-Gudzinas 2026m). The Foundations record argues that Shor's algorithm is a theoretical artifact defined under unrealizable conditions, and that error correction fails to suppress errors beyond current scales due to correlated failures (Quni-Gudzinas 2026i).

The program's own archive is imperfect, and this paper records the imperfections rather than hiding them. Two pairs of records are duplicate or near-duplicate deposits rather than clean version chains (the Spin-Free Substrate pair (Quni-Gudzinas 2026p, 2026q) and the vortex-feasibility pair (Quni-Gudzinas 2026u, 2026v)); version labels across the set are inconsistent; and the archive's internal identifier table carries missing and cross-wired DOI fields for several records (Section 12). A converged program should have a converged archive; the disorder is disclosed here as a data-quality finding.

3. Lineage: How the Register Descends from Earlier Work

The sixteen records did not appear from nowhere. Each movement is the latest link in a published chain, and the chain is the evidence that this paper is a continuation rather than a restart. A reader can follow the citations backward from any register entry to its oldest published premise.

The audit lineage. The critique of quantum-computing claims began with the reassessment of quantum computation's foundations (Quni-Gudzinas 2026i), proceeded through the energy-accounting benchmarks of the competitive landscape (QNFO Research 2026b), the qudit extension (Quni-Gudzinas 2026o), and the quantum-neural-network audit (QNFO Research Collective and Quni-Gudzinas 2026), and reaches the trapped-ion due-diligence record cited in this paper (Quni-Gudzinas 2026t). The register's energy discipline is inherited from this chain.

The ultrametric lineage. The claim that p-adic structure is physically testable descends from the number-theoretic classification of error-correcting codes (QNFO Research 2026c), the ultrametric bridge theorem (Quni-Gudzinas 2026r), and the conditional-state analysis that turned the theorem into a measurable prediction (Quni-Gudzinas 2026c) before the trapped-ion protocol existed (QNFO Research and Quni-Gudzinas 2026a). The Zitterbewegung entry of the register (QNFO Research and Quni-Gudzinas 2026b) continues the earlier analysis of the effect's ultrametric readout and its bridge to anyon braiding (Quni-Gudzinas 2026w). The anomalous-diffusion entry (Quni-Gudzinas 2026j) descends from the dimensionless reformulation program (Quni-Gudzinas 2026n; Ostrowski 1916), the non-anthropocentric units reformulation (Quni-Gudzinas 2026f), and the continuum analysis that bounds which real-number structures carry physical content (Quni-Gudzinas 2026d).

The architectural lineage. The passive-design claim descends from the architectonics critique of active error correction (Quni-Gudzinas 2026h), the spin-free substrate protocol (Quni-Gudzinas 2026p), and the autonomous GKP stabilization proposal (Quni-Gudzinas 2025), and is corroborated by the ultrametric metrology proposal (Quni-Gudzinas 2026g). The dissipation-first reading of the tree hierarchy is this paper's own synthesis of those published claims; the chain is cited so each link can be audited.

The methodological lineage. The falsifiability discipline applied here descends from the falsifiability-crisis analysis (Quni-Gudzinas 2026l), the reification analysis (Quni-Gudzinas 2026m), the falsifiability protocol for discrete-continuum signatures (Quni-Gudzinas 2026a), and the fifteen-question ignorance audit with its companion case study (Quni-Gudzinas 2026s, 2026e). The register's kill-conditions are that method applied to the program's own claims.

The consilience lineage. The synthesis of number theory and physics into one tree-shaped structure was published in the five-pillar consilience paper (QNFO Research 2026a) and the adelic core synthesis (Quni-Gudzinas 2026b). The present paper extends that line from structure to instrumentation: where the predecessor papers established the shared geometry, this paper supplies the devices that measure it.

What the chain shows: the three movements share a published ancestry, and each movement's latest record cites the earlier links. The chain of reasoning from the oldest premise (the completions of the rationals (Ostrowski 1916)) to the newest instrument (a register of five kill-conditions, Section 4) runs through every one of the sixteen records.

4. The Testability Movement: A Register of Five Falsifiable Observables

The register is the paper's core artifact. Each entry states the observable, the pre-registered prediction, the kill-condition, and the apparatus.

R1 – Ultrametricity violation rate in a Page-Wootters clock. Observable: the Parisi ultrametricity violation rate (UVR) of conditional state overlaps, measured by conditional state tomography on a single trapped ion with electronic states as the clock and motional Fock states as the rest. Prediction: UVR equal to zero for diagonal clock-rest coupling; UVR in the 29-35 percent band for nondiagonal coupling (QNFO Research and Quni-Gudzinas 2026a; Quni-Gudzinas 2026c). Kill-condition: a measured UVR indistinguishable between the two coupling classes – the theory predicts the split, and the split is what is tested. Apparatus: existing trapped-ion capability; estimated eight weeks of beam time.

R2 – The 2-adic Zitterbewegung frequency. Observable: the Zitterbewegung frequency ratio in a Dirac-simulator implementation. Prediction: $\sqrt{2} \cdot 2mc^2/\hbar$, a 41.4 percent deviation from the Archimedean value (QNFO Research and Quni-Gudzinas 2026b). Kill-condition: a measured ratio consistent with 1 within error. Open constraint: the vanishing-signal question – whether the ZBW signal is observable at all in the Foldy-Wouthuysen frame – and the amplification-feasibility question remain unresolved in the source records (Quni-Gudzinas 2026u, 2026v); R2 is therefore registered as a measurement whose SNR budget must be closed before the experiment is meaningful, not as a settled test.

R3 – Effective transient dimension on p-regular trees. Observable: the return probability decay of a random walk on p-adic trees, extractable in simulation and, in principle, in engineered hierarchical coupling graphs. Prediction: effective transient dimensions near 6.2 (p=2) and 7.9 (p=3), against the Archimedean baseline near 1 (Quni-Gudzinas 2026j). Kill-condition: dimension growth that flattens with p. This is the register's simulation-first entry: it is executable today, with the code already deposited with the source record.

R4 – Dissipative break-even. Observable: logical bit-flip suppression of an autonomously stabilized GKP state as a function of the cooling-to-heating rate ratio. Prediction: convergence to the GKP manifold with exponential bit-flip suppression once the ratio exceeds π (Quni-Gudzinas 2025). Kill-condition: no break-even at the predicted ratio in any ion species. Constraint: break-even on one mode is not a logical qubit; the fault-tolerance step remains unclaimed.

R5 – Tensor-network collapse under nonlocal perturbations. Observable: whether the classical simulation advantage of tensor networks for local Hamiltonian dynamics persists when the Hamiltonian acquires controlled nonlocal terms. The tensor-network record reads the recent tensor-network simulation results as evidence that local-Hamiltonian dynamics are ultrametric, because local structure is exactly where the Bruhat-Tits hierarchy lives (Quni-Gudzinas 2026k). This is the register's retrodiction entry, graded honestly in the Bayesian accounting (Section 6): the standard area-law explanation predicts the same success, so R5 is registered as a discriminating test rather than credited as evidence. Prediction under the ultrametric reading: the advantage degrades under nonlocal perturbation faster than area-law extrapolation would predict. Kill-condition: advantage persists at area-law-predicted levels. This entry exists precisely because the tensor-network reading currently carries zero independent evidential weight and needs a test to earn any.

5. Negative Results the Register Must Respect

A testable program is one that keeps its negative results. Five are load-bearing here.

First, generic clock-rest coupling does NOT produce ultrametricity. The 8,000-system study found a 29-35 percent violation rate in the generic case (Quni-Gudzinas 2026c). This matters because an earlier, naive version of the program's expectation – that ultrametricity would emerge generically from any Page-Wootters construction – is thereby ruled out. The surviving claim is the sharp sufficient condition: diagonal coupling. The trapped-ion protocol is designed around exactly that surviving claim. A reader who only sees the positive claim without the ruled-out generic case would misprice the theory.

Second, trapped ions are the wrong economics play. The joules-per-solution competitive landscape ranks trapped ions last among seventeen platforms (8.5-16.3 J/sol) because 50-100 microsecond gate times overwhelm their room-temperature power advantage (QNFO Research 2026b). The testbed movement is therefore a physics claim, not a production claim; the architectural movement (passive stabilization) exists in part to attack the gate-time problem that produces this ranking. The qudit extension projects roughly 10^{-5} J/sol for a p-adic-coded qudit platform and pre-registers its own disconfirmation condition (Quni-Gudzinas 2026o) – a projection, not a measurement.

Third, the ZBW signal may not be observable at all in the natural readout frames, and the amplification question is open. R2 is registered with this constraint attached, not buried.

Fourth, the audit discipline has produced verdicts against the program's own preferred direction before: the quantum-neural-network audit sustained the methodology claim and rejected the advantage claim (QNFO Research Collective and Quni-Gudzinas 2026). The register inherits that willingness.

Fifth, error correction at scale is diagnosed as failing beyond current scales due to correlated failures rather than independent noise (Quni-Gudzinas 2026i) – the constraint that motivates the passive path in the first place.

6. Evidential Weight: What Carries Weight and What Does Not

Under a Bayesian accounting, the register's entries carry unequal weight.

R1 and R2 are pre-registered predictions: their values were published before measurement (QNFO Research and Quni-Gudzinas 2026a, 2026b). A measured match would be surprising under the null hypothesis of no ultrametric structure, so both carry potential positive evidential weight – and none until measured. R4 is likewise pre-registered (Quni-Gudzinas 2025). R3 is simulation evidence: it constrains the modeling claim, not the physics claim.

The tensor-network reading (R5's target) carries ZERO current evidential weight. The area-law explanation of tensor-network success is the incumbent, and it predicts the observed success with no reference to p-adic structure; the ultrametric reading was formulated after the results it explains. It is labeled here as retrodiction, and it is included in the register only as a test to be run, not as evidence in hand (Quni-Gudzinas 2026k).

The falsifiability discipline applies symmetrically. The records that diagnose unfalsifiability in incumbent frameworks (Quni-Gudzinas 2026l) are the same standard under which the program's own claims are graded here: R5 is the demonstration that the standard is applied inward.

7. The Architectural Movement: Passive Design as the Embodiment

The architectural claim is that the tree hierarchy is not only a geometry to be measured but a design principle to be built. Three records carry the argument.

The GKP stabilization record replaces measurement-based feedback with engineered dissipative reservoir dynamics: a mixed-species crystal couples the logical mode to a coolant ion, exporting entropy continuously, with convergence to the GKP manifold when cooling beats heating by more than π (Quni-Gudzinas 2025). This is a hierarchy-respecting design in a precise sense: stabilization is delegated to the environment's own structure rather than imposed by an external controller – the passive reading of the tree.

The Spin-Free Substrate protocol is the architectural bridge to simulation: it maps Posner-molecule nuclear spin dynamics (J-couplings of 0.003-0.178 Hz; dipolar relaxation treated as a Lindblad channel) onto 4-qubit trapped-ion circuits requiring six global Molmer-Sorensen gates per simulated second at 96.9 percent fidelity (Quni-Gudzinas 2026p). The design principle named there – the spin-free substrate – is a MAP label: a design pattern, not a new entity, and this paper keeps it at that status.

The metrology record extends the passive claim to sensing: hierarchical noise correlations exploited for sub-shot-noise precision without active correction, claimed on a room-temperature photonic demonstration (Quni-Gudzinas 2026g). The demonstration claim is reported as claimed by the source record; independent replication is not yet on record.

The architectonics record supplies the thermodynamic framing: active correction fights the second law and pays energy for the fight; passive structure works with it (Quni-Gudzinas 2026h). The dimensionless reformulation supplies the bookkeeping that keeps the program's formulas place-democratic (Quni-Gudzinas 2026n).

8. The Methodological Movement: Discipline as Infrastructure

The third movement is the reason the first two do not float away. Its three records name the failure modes: five structural patterns of unfalsifiability (Quni-Gudzinas 2026l); the reification pattern by which models become mistaken for objects (Quni-Gudzinas 2026m); and the theoretical-artifact pattern by which idealized algorithms misdirect engineering programs (Quni-Gudzinas 2026i).

Two instruments operationalize the discipline. The Bayesian delta-log-odds gate assigns zero weight to explanations built after the observations they explain – applied inward in Section 6 to the program's own tensor-network reading. The joules-per-solution criterion turns advantage claims into energy accounting, applied in the audit family (QNFO Research Collective and Quni-Gudzinas 2026; QNFO Research 2026b; Quni-Gudzinas 2026o). A practitioner can adopt either instrument without adopting any of the program's physics.

9. Practitioner Section: What Can Be Built Today

Nothing in this paper requires new theory to become useful. Four artifacts can be built by an engineering team using the cited records alone.

Artifact 1 – The decision-tool register (spec-sheet). Section 4 IS the spec-sheet. An ion-trap team can take R1 directly to an experiment proposal: single ion, carrier and sideband transitions, conditional state tomography, eight-week estimate, with the predicted UVR split (0 percent diagonal versus 29-35 percent nondiagonal) as the accept/reject criterion. The domain of validity is conditional: R1 holds for clock-rest coupling engineered as specified; it says nothing about platforms without clock structure. A superconducting team gains nothing from R1; a neutral-atom team might, if a clock degree of freedom is available – that is the conditional truth, stated plainly.

Artifact 2 – The SNR budget template for R2. The ZBW test cannot be proposed until the signal-to-noise question is closed. The two feasibility records contain the structure of the budget (amplification mechanism, areal-rate observable, noise floor); a team can turn them into a spreadsheet decision tool: parameterize amplification gain, measurement time, and heating rate, and compute whether $\sqrt{2} \cdot 2mc^2/\hbar$ clears the floor on available hardware (Quni-Gudzinas 2026u, 2026v). The tool IS the deliverable; the experiment is downstream of it.

Artifact 3 – The reproducible simulation kit. R3 is executable today: the Monte-Carlo code for anomalous diffusion on p-regular trees is deposited with the source record (Quni-Gudzinas 2026j). A team can re-run, extend to p equal to 5 and 7, and publish the extended dimension table as a benchmark artifact. No hardware required.

Artifact 4 – The energy-audit template. The joules-per-solution methodology is fully specified across the competitive-landscape records (QNFO Research 2026b; Quni-Gudzinas 2026o). Any hardware team can apply it to its own platform: the metric, the conservative-bound discipline, and the published baseline values are all in the cited records. This artifact works regardless of whether any ultrametric claim survives.

The practitioner-facing summary is one sentence: the program hands out one ready experiment (R1), one spreadsheet problem (R2), one benchmark (R3), and one audit template (Artifact 4) – each usable without subscribing to the interpretation that motivated it.

10. Premise-Depth Disclosure

The derived content of this paper is organizational and arithmetic: the register, the resource counts, the predicted values, and the evidential-weight grading. The physical content is imported, and the imports are named.

Import 1: the Ultrametric Bridge Theorem – conditional quantum states under a global constraint necessarily form ultrametric hierarchies, with hierarchy depth fixing the radix (Quni-Gudzinas 2026r). Import 2: the Silent Parameter Principle – the truncated SU(2) representation ring is isomorphic to the character ring of the 2-adic units, from which the $\sqrt{2}$ Zitterbewegung frequency follows (QNFO Research and Quni-Gudzinas 2026b). Supporting imports: Ostrowski's theorem as the place-democracy frame (Ostrowski 1916; Quni-Gudzinas 2026n), and the Page-Wootters conditional-state formalism (Page and Wootters 1983).

Where the premises END: R1's predictions end at the Bridge Theorem's sufficient condition. R2's predicted value ends at the Silent Parameter Principle. R5's reading ends at the identification of local-Hamiltonian dynamics with tree structure – a MAP claim, not a derived one. R3 and R4 are simulation and engineering claims whose premises end at the stated physical models (regular-tree random walks; Lindblad dissipative dynamics). No claim in this paper reaches below these inputs. A reader who rejects Import 1 can discard R1 without touching the rest of the register; a reader who rejects Import 2 can discard R2. That separability is deliberate: the register is designed so that its entries fail independently.

11. Falsifiability Conditions

Formally, the paper is disconfirmed by any of the following measurements:

1. UVR statistically indistinguishable between diagonal and nondiagonal clock-rest coupling (kills R1 and, with it, the testable bearing of the Bridge Theorem).
2. ZBW frequency ratio consistent with 1 (kills R2).
3. Effective transient dimension growth flattening with p (kills R3's fractal-trap reading).
4. No dissipative break-even at cooling-to-heating ratio exceeding π in any ion species (kills R4).
5. Tensor-network advantage persisting at area-law-predicted levels under nonlocal perturbations (kills R5's ultrametric reading and confirms the incumbent).

A world in which all five conditions obtain is a world in which the sixteen records are interesting but disconnected engineering notes, and this paper's organizational claim – that they are one program – is the thing that fails. The paper asserts no stronger consequence.

12. Data-Quality Findings

Cross-system identifier audit of the input records surfaced seven findings: the two vortex-feasibility records are cross-wired in the archive's identifier table (each row carrying the other's DOI); three records lack DOI fields entirely; one record carries a generic internal title; and an identifier-type field is systematically mislabeled. The duplicate-record pairs (Section 2) are the archive-level expression of the same disorder. These findings are published here so that the archive can be repaired, and because a testable program that hides its bookkeeping errors invites the reification failure its own methodology record diagnoses (Quni-Gudzinas 2026m).

13. Conclusion

Sixteen records, three movements, one testable structure. The contribution is not the claim that ultrametric structure exists in quantum dynamics; it is the claim that the structure is now cheap to test, and the register that makes the tests concrete. R1 is an eight-week experiment on existing apparatus. R3 runs on a laptop. The architectural movement gives the design language; the methodological movement gives the discipline; the register gives the practitioners something to build, measure, or reject. If the register fills with null results, the program has still contributed: five kill-conditions executed is five pieces of knowledge gained, and the falsifiability discipline the records advocate will have been demonstrated on their own claims.

References

Hrmo, Pavel, Benjamin Wilhelm, Lukas Gerster, et al. 2023. "Native Qudit Entanglement in a Trapped Ion Quantum Processor." *Nature Communications* 14: 2242. https://doi.org/10.1038/s41467-023-37375-2.

Meth, Michael, Jan Frederik Haase, Jinglei Zhang, et al. 2024. "Digital Quantum Simulation of a (1+1)d SU(2) Lattice Gauge Theory with Ion Qudits." *PRX Quantum* 5: 040309. https://doi.org/10.1103/PRXQuantum.5.040309.

Ostrowski, Alexander. 1916. "Über Einige lösungen Der Funktionalgleichung $\varphi(x)\varphi(y) = \varphi(xy)$." In *Acta Mathematica*, vol. 41.

Page, Don N., and William K. Wootters. 1983. "Evolution Without Evolution: Dynamics Described by Stationary Observables." *Physical Review D* 27 (12): 2885–92. https://doi.org/10.1103/PhysRevD.27.2885.

QNFO Research. 2026a. *Five Pillars, One Structure: Consilient Convergence in QNFO Research*. Zenodo. https://doi.org/10.5281/zenodo.21547793.

QNFO Research. 2026b. *JPCUB Competitive Landscape V2.0: System-Level Joules-Per-Solution Estimates for 17 Quantum Computing Platforms from Published Specifications*. Zenodo. https://doi.org/10.5281/zenodo.21821767.

QNFO Research. 2026c. *Number-Theoretic Ultrametric Foundations: A Unified p-Adic Framework for Error-Correcting Code Classification*. Zenodo. https://doi.org/10.5281/zenodo.21193487.

QNFO Research Collective, and Rowan Brad Quni-Gudzinas. 2026. *Auditing the BQNN: Does a Tunable Quantum Neural Network on Trapped-Ion and Superconducting Hardware Demonstrate a Route to Near-Term Quantum Advantage?* Zenodo. https://doi.org/10.5281/zenodo.21566035.

QNFO Research, and Rowan Brad Quni-Gudzinas. 2026a. *Trapped-Ion Page-Wootters Experiment: Protocol for Testing Ultrametricity*. Zenodo. https://doi.org/10.5281/zenodo.21120469.

QNFO Research, and Rowan Brad Quni-Gudzinas. 2026b. *Zitterbewegung: From Archimedean Puzzle to Adelic Observable*. Zenodo. https://doi.org/10.5281/zenodo.21600628.

Quni-Gudzinas, Rowan Brad. 2025. *Autonomous Dissipative Stabilization of Gottesman-Kitaev-Preskill States in Room-Temperature Ion Traps*. Zenodo. https://doi.org/10.5281/zenodo.17794233.

Quni-Gudzinas, Rowan Brad. 2026a. *A Falsifiability Protocol for the q-Fundamental Hypothesis: Experimental Signatures of the Discrete Continuum*. Zenodo. https://doi.org/10.5281/zenodo.21697717.

Quni-Gudzinas, Rowan Brad. 2026b. *Adelic Core Synthesis: Cross-Domain Foundations of Adelic QFT*. Zenodo. https://doi.org/10.5281/zenodo.21786473.

Quni-Gudzinas, Rowan Brad. 2026c. *Conditional State Distances in Page-Wootters Quantum Clocks: When Does Ultrametricity Emerge?* Zenodo. https://doi.org/10.5281/zenodo.21120286.

Quni-Gudzinas, Rowan Brad. 2026d. *Continuum Trilogy: Papers i-III – the Ontological Closure Program*. Zenodo.

https://doi.org/10.5281/zenodo.21672990.
Quni-Gudzinas, Rowan Brad. 2026e. *Knowing What We Do Not Know: Ignorance Auditing, AI-Generation Detection, and the Epistemic Lessons of an AI-Assisted Research Pipeline*. Zenodo. https://doi.org/10.5281/zenodo.21901983.
Quni-Gudzinas, Rowan Brad. 2026f. *Non-Anthropocentric Natural Units: From the Bekenstein Bound to Ostrowski's Theorem*. Zenodo. https://doi.org/10.5281/zenodo.21480756.
Quni-Gudzinas, Rowan Brad. 2026g. *Passive Error Resilience Through Ultrametric Geometry: A Proposal for p-Adic Quantum Metrology*. Zenodo. https://doi.org/10.5281/zenodo.21748299.
Quni-Gudzinas, Rowan Brad. 2026h. *Quantum Architectonics: The Passive Path to Scale*. Zenodo. https://doi.org/10.5281/zenodo.18522367.
Quni-Gudzinas, Rowan Brad. 2026i. *Reassessing the Foundations of Quantum Computation: From Theoretical Artifacts to Physical Realities*. Zenodo. https://doi.org/10.5281/zenodo.21993655.
Quni-Gudzinas, Rowan Brad. 2026j. *Spectral Analysis of Anomalous Diffusion on p-Adic Fractals*. Zenodo. https://doi.org/10.5281/zenodo.18606514.
Quni-Gudzinas, Rowan Brad. 2026k. *Tensor Networks as Bruhat-Tits Tree Computation*. Zenodo. https://doi.org/10.5281/zenodo.21820137.
Quni-Gudzinas, Rowan Brad. 2026l. *The Falsifiability Crisis in Contemporary Physics: LCDM, the Standard Model, and the Measurement-Exercise Trap*. Zenodo. https://doi.org/10.5281/zenodo.21791457.
Quni-Gudzinas, Rowan Brad. 2026m. *The Meta-Pattern of Reification in Physics*. Zenodo. https://doi.org/10.5281/zenodo.19605446.
Quni-Gudzinas, Rowan Brad. 2026n. *The Ostrowski Dimensionless Reformulation: A Systematic Compilation of Fundamental Physics Equations in Planck Units*. Zenodo. https://doi.org/10.5281/zenodo.21754024.
Quni-Gudzinas, Rowan Brad. 2026o. *The Qudit Advantage: System-Level Joules-Per-Solution Comparison of a Qudit Architecture Against 17 Conventional Qubit Quantum Computing Platforms*. Zenodo. https://doi.org/10.5281/zenodo.21827737.
Quni-Gudzinas, Rowan Brad. 2026p. *The Spin-Free Substrate: Trapped-Ion Quantum Simulation of Posner Molecule Nuclear Spin Dynamics*. Zenodo. https://doi.org/10.5281/zenodo.20411697.
Quni-Gudzinas, Rowan Brad. 2026q. *The Spin-Free Substrate: Trapped-Ion Quantum Simulation of Posner Molecule Nuclear Spin Dynamics (Companion Record)*. Zenodo. https://doi.org/10.5281/zenodo.20411734.
Quni-Gudzinas, Rowan Brad. 2026r. *The Ultrametric Bridge Theorem: Conditional Quantum States Under Global Constraints Necessarily Form Ultrametric Hierarchies*. Zenodo. https://doi.org/10.5281/zenodo.21102770.
Quni-Gudzinas, Rowan Brad. 2026s. *The Universal Ignorance Audit: A Fifteen-Question Method for Systematic Inquiry into the Structure of Not-Knowing*. Zenodo. https://doi.org/10.5281/zenodo.21901984.
Quni-Gudzinas, Rowan Brad. 2026t. *Trapped-Ion Qudit Quantum Computing: Due-Diligence Assessment of the Ringbauer Program*. Zenodo. https://doi.org/10.5281/zenodo.21879231.
Quni-Gudzinas, Rowan Brad. 2026u. *Vortex-Enhanced Zitterbewegung: Amplification Feasibility for Trapped-Ion Dirac Simulators*. Zenodo. https://doi.org/10.5281/zenodo.21221899.
Quni-Gudzinas, Rowan Brad. 2026v. *Vortex-Enhanced Zitterbewegung: Amplification Feasibility for Trapped-Ion Systems*. Zenodo. https://doi.org/10.5281/zenodo.21336123.

Quni-Gudzinas, Rowan Brad. 2026w. *Zitterbewegung as the Physical Realization of p-Adic Anyon Braiding*. Zenodo. https://doi.org/10.5281/zenodo.21214362.

Ringbauer, Martin, Michael Meth, Lukas Postler, et al. 2022. "A Universal Qudit Quantum Processor with Trapped Ions." *Nature Physics* 18: 1053–57. https://doi.org/10.1038/s41567-022-01658-0.